User Stories

CareQueue - Hospital Digital Token and Queue Management System

Field	Value
Document	User Stories with Acceptance Criteria (Product Backlog, part 2 of 2)
Product	CareQueue - Hospital Digital Token and Queue Management System
Problem statement	Patients wait in queues for registration and billing. Build a digital token and queue-management system that displays the current token and estimated waiting time.
Traces to	CareQueue MRD v1.0 and CareQueue PRD v1.0
Version / Status	v1.0 - Draft for hackathon submission
Date	24 October 2026
Team	Future Coders
Roles	BA: Sam Austin Dev: Ashik Rasool UI/UX Designer: Nishandhini QA Tester: Rogasree

1. Story Summary

Priority comes from the PRD (MoSCoW). Size: S = under 2 hours, M = 2-4 hours, L = more than 4 hours. Acceptance criteria are written as Given / When / Then and numbered so QA can mark each one pass or fail.

ID	Story	Epic	Priority	PRD	Size
US-01	Issue a token for the right queue	E1	Must	M1	M
US-02	Number tokens in order, reset daily	E1	Must	M1	S
US-03	Refuse missing or invalid details	E1	Must	M1	S
US-04	Send a patient on to billing	E1	Should	S4	M
US-05	Call the next patient with one click	E2	Must	M2	S
US-06	Show now-serving on the waiting-room display	E2	Must	M2	M
US-07	Skip or recall an absent patient	E2	Should	S1	S
US-08	Announce the called token	E2	Should	S2	S
US-09	Show an estimated wait for every token	E3	Must	M3	M
US-10	Learn from real service times	E3	Must	M3	M
US-11	Serve from more than one counter	E3	Should	S5	L
US-12	Print or download the token slip	E4	Could	M1	S
US-13	Track my token on my phone	E4	Should	S3	M
US-14	Sign in as patient or receptionist	E5	Should	S6	M
US-15	See today's queue summary	E6	Could	C2	M
US-16	Find the busiest hour and export the day	E6	Could	C2	S

E1 - Token issue

US-01 Issue a token for the right queue

As a receptionist, **I want** to enter a patient's name, choose Registration or Billing and issue a token, **so that** every patient gets a clear place in the right queue.

Priority: **Must** | Epic: E1 | PRD: M1 | MRD: MR-1 | Size: M

AC	Given	When	Then
AC-01.1	the staff screen is open	I enter a name, choose Registration and press Issue Token	a token such as R-001 is created and shown on screen with its position in the queue
AC-01.2	the staff screen is open	I choose Billing instead	the token is created in the billing queue with the B prefix (e.g. B-001)
AC-01.3	I am issuing tokens during a normal day	I issue one token	it takes one form submit and no more than 15 seconds from opening the form

US-02 Number tokens in order, reset daily

As a receptionist, **I want** tokens in each queue to be numbered in order and to start again each day, **so that** patients can see the order is fair and numbers stay short.

Priority: **Must** | Epic: E1 | PRD: M1 | MRD: MR-1 | Size: S

AC	Given	When	Then
AC-02.1	R-004 was the last registration token today	I issue a new registration token	it is R-005; billing numbering is unaffected
AC-02.2	tokens were issued yesterday	I issue the first token today	numbering restarts at R-001 / B-001
AC-02.3	many tokens have been issued today	I look at all of them	no token number appears twice in the same queue on the same day

US-03 Refuse missing or invalid details

As a receptionist, **I want** the form to stop me when a detail is missing or wrong, **so that** no token is issued with bad data.

Priority: **Must** | Epic: E1 | PRD: M1 | MRD: MR-1 | Size: S

AC	Given	When	Then
AC-03.1	the name field is empty or no queue is chosen	I press Issue Token	no token is created and each problem field shows a message saying what to fix
AC-03.2	I enter an optional phone number	it is not 10 digits starting with 6-9	the form shows an error for the phone field and does not issue a token
AC-03.3	I fix every error	I press Issue Token again	the token is issued and the error messages disappear

US-04 Send a patient on to billing

As a receptionist, **I want** to move a patient who has finished registration into the billing queue with one click, **so that** they keep their place in the flow without re-entering details.

Priority: **Should** | Epic: E1 | PRD: S4 | MRD: MR-1 | Size: M

AC	Given	When	Then
AC-04.1	R-012 is being served at registration	I press Send to Billing	a new billing token (e.g. B-007) is created for the same patient and R-012 is marked done
AC-04.2	a billing token was created this way	I look at it in the staff screen	it shows the linked registration token number

E2 - Calling and Now-Serving display

US-05 Call the next patient with one click

As a receptionist, **I want** to press Call Next to call the next waiting token, **so that** I don't have to shout names or manage the crowd.

Priority: **Must** | Epic: E2 | PRD: M2 | MRD: MR-2 | Size: S

AC	Given	When	Then
AC-05.1	tokens are waiting in a queue	I press Call Next	the first waiting token becomes 'Now serving' and the previous one is marked done
AC-05.2	no tokens are waiting	I press Call Next	nothing changes and the screen says 'No one is waiting'
AC-05.3	a token was called	I look at the waiting list	the called token is no longer listed as waiting

US-06 Show now-serving on the waiting-room display

As a patient in the waiting area, **I want** to see the token being served and the next tokens on a big screen, **so that** I know how close my turn is without standing at the counter.

Priority: **Must** | Epic: E2 | PRD: M2 | MRD: MR-2 | Size: M

AC	Given	When	Then
AC-06.1	the display is open on a second screen	the receptionist presses Call Next	the display shows the new token as Now Serving within 2 seconds, without a page refresh
AC-06.2	tokens are waiting	I look at the display	it shows the next 5 waiting tokens for each queue in order
AC-06.3	tokens were issued with patient names	I look at the display	only token numbers are shown, never names
AC-06.4	the display is viewed from 6 metres	I read the Now Serving token	the token number is at least 120 px high and readable

US-07 Skip or recall an absent patient

As a receptionist, **I want** to skip a patient who does not come and recall them later, **so that** one absent patient does not block the line.

Priority: **Should** | Epic: E2 | PRD: S1 | MRD: MR-2 | Size: S

AC	Given	When	Then
AC-07.1	the called patient has not come	I press Skip	the token moves to a Missed list and the next token can be called
AC-07.2	a token is in the Missed list	I press Recall on it	it becomes Now Serving again and the display updates

US-08 Announce the called token

As a elderly patient, **I want** to hear my token number announced when it is called, **so that** I don't miss my turn if I can't see the screen.

Priority: **Should** | Epic: E2 | PRD: S2 | MRD: MR-2 | Size: S

AC	Given	When	Then
AC-08.1	the display has sound on	a token is called	a chime plays and a voice says, for example, 'Token R 0 1 2, please come to Registration'
AC-08.2	a token is recalled	it becomes Now Serving again	the announcement plays again

E3 - Waiting-time estimate

US-09 Show an estimated wait for every token

As a patient, **I want** to see my estimated waiting time next to my token, **so that** I can decide to sit down or step out.

Priority: **Must** | Epic: E3 | PRD: M3 | MRD: MR-3 | Size: M

AC	Given	When	Then
AC-09.1	AST is 4 min, 1 counter open, the current patient has been served for 1 min and 5 tokens are ahead	the estimate is calculated	it shows 23 min, using EWT = R + N x AST, rounded up to the minute
AC-09.2	tokens are waiting	any queue event happens (issue, call, skip, complete)	every waiting token's estimate is recalculated and shown on the staff screen and display
AC-09.3	a token is Now Serving	I look at it	it shows 'Now serving', not a waiting time

US-10 Learn from real service times

As a receptionist, **I want** the estimate to use how fast we really serve patients today, **so that** patients get honest numbers instead of a fixed guess.

Priority: **Must** | Epic: E3 | PRD: M3 | MRD: MR-3 | Size: M

AC	Given	When	Then
AC-10.1	fewer than 3 tokens have been served in a queue today	the estimate is calculated	it uses the default service time of 3 minutes
AC-10.2	3 or more tokens have been served	the estimate is calculated	AST is the average of the last 10 served tokens (time from Call to done) in that queue
AC-10.3	the actual waits are compared with the estimates after the first 3 tokens	QA reviews a test run	80% of tokens are within +/- 5 minutes of their estimate (PRD metric)

US-11 Serve from more than one counter

As a receptionist, **I want** to open up to 3 counters for a queue, each calling its own patient, and have the estimate know how many are open, **so that** two of us can serve at once and waits are not overstated.

Priority: **Should** | Epic: E3 | PRD: S5 | MRD: MR-3 | Size: L

AC	Given	When	Then
AC-11.1	2 counters are open for a queue	the estimate is calculated	N is divided by 2 in the formula
AC-11.2	a counter is paused	the estimate is recalculated	the paused counter is not counted, and it cannot call patients until it is opened again
AC-11.3	Counter 1 is serving G004	the receptionist at Counter 2 presses Call Next	G005 is called to Counter 2 and G004 stays at Counter 1
AC-11.4	a token is called to a counter	the display, voice announcement and patient's phone update	each names the counter, e.g. 'G005, Counter 2'

E4 - Patient's token copy

US-12 Print or download the token slip

As a patient, **I want** a copy of my token on paper or on my phone, **so that** I can keep my number with me.

Priority: **Could** | Epic: E4 | PRD: M1 | MRD: MR-1 | Size: S

AC	Given	When	Then
AC-12.1	a token has been issued	the receptionist presses Print	one page prints with the token number, queue, time issued and estimated wait
AC-12.2	a token has been issued	the receptionist presses Download	an image of the same slip is saved

US-13 Track my token on my phone

As a patient, **I want** to scan a QR code on my token and see my position live, **so that** I don't have to keep watching the screen.

Priority: **Should** | Epic: E4 | PRD: S3 | MRD: MR-3 | Size: M

AC	Given	When	Then
AC-13.1	I scan the QR code on my slip	the page opens	it shows my token, the token being served, people ahead and my estimated wait, with no app and no login
AC-13.2	I keep the page open	the queue moves	the page updates without refresh
AC-13.3	2 or fewer patients are ahead of me	the page updates	it shows an 'Almost your turn' alert

E5 - Sign-in and access

US-14 Sign in as patient or receptionist

As a patient or receptionist, **I want** to choose my role on a landing page and sign in, **so that** I reach the right screen and patients cannot open the staff console.

Priority: **Should** | Epic: E5 | PRD: S6 | MRD: MR-1, MR-2 | Size: M

AC	Given	When	Then
AC-14.1	the landing page is open	I look at it	I can choose Patient or Receptionist, and no sign-in form shows until I choose
AC-14.2	I chose Patient	I enter my name and a 10-digit phone number and continue	the token page opens with my name already filled in
AC-14.3	I chose Receptionist	I enter the PIN (1234 in the demo)	the staff console opens; a wrong PIN shows an error and keeps me on the landing page
AC-14.4	nobody is signed in	I open the token page or the staff console directly	I am sent back to the landing page

E6 - Queue reports

US-15 See today's queue summary

As a hospital administrator, **I want** to see today's tokens, waits and missed patients for every department on one page, **so that** I can tell how the queue is performing without counting slips by hand.

Priority: **Could** | Epic: E6 | PRD: C2 | MRD: MR-3 | Size: M

AC	Given	When	Then
AC-15.1	tokens were issued and served today	I open the Reports page	each department shows tokens issued, served, waiting and missed, and the totals match the staff console
AC-15.2	3 tokens in a department have been called and completed	the report is calculated	it shows the average wait (issued to called) and the average service time (called to done) in minutes
AC-15.3	the Reports page is open	a token is issued or called	the numbers update within 2 seconds without a page refresh
AC-15.4	nobody is signed in as staff	I open the Reports page	I am sent back to the landing page

US-16 Find the busiest hour and export the day

As a hospital administrator, **I want** to see how many tokens were issued in each hour and download the day's figures, **so that** I can put more staff on at the busiest time and keep a record.

Priority: **Could** | Epic: E6 | PRD: C2 | MRD: MR-3, MR-4 | Size: S

AC	Given	When	Then
AC-16.1	tokens were issued at different hours today	I open the Reports page	a bar chart shows tokens issued per hour and the busiest hour is highlighted and named
AC-16.2	the report is on screen	I press Download CSV	a CSV file with one row per department and a total row is saved, with the same numbers and no patient names or phone numbers

Traceability Matrix

Story	Epic	PRD feature	MRD need	No. of ACs
US-01	E1	M1	MR-1	3
US-02	E1	M1	MR-1	3
US-03	E1	M1	MR-1	3
US-04	E1	S4	MR-1	2
US-05	E2	M2	MR-2	3
US-06	E2	M2	MR-2	4
US-07	E2	S1	MR-2	2
US-08	E2	S2	MR-2	2

Story	Epic	PRD feature	MRD need	No. of ACs
US-09	E3	M3	MR-3	3
US-10	E3	M3	MR-3	3
US-11	E3	S5	MR-3	4
US-12	E4	M1	MR-1	2
US-13	E4	S3	MR-3	3
US-14	E5	S6	MR-1, MR-2	4
US-15	E6	C2	MR-3	4
US-16	E6	C2	MR-3, MR-4	2

Test Checklist (for QA)

QA (Rogasree) marks each acceptance criterion as the story lands. This table becomes the submitted test checklist.

AC	Story	Priority	Pass / Fail	Notes
AC-01.1	US-01	Must		
AC-01.2	US-01	Must		
AC-01.3	US-01	Must		
AC-02.1	US-02	Must		
AC-02.2	US-02	Must		
AC-02.3	US-02	Must		
AC-03.1	US-03	Must		
AC-03.2	US-03	Must		
AC-03.3	US-03	Must		
AC-04.1	US-04	Should		
AC-04.2	US-04	Should		
AC-05.1	US-05	Must		
AC-05.2	US-05	Must		
AC-05.3	US-05	Must		
AC-06.1	US-06	Must		
AC-06.2	US-06	Must		
AC-06.3	US-06	Must		
AC-06.4	US-06	Must		
AC-07.1	US-07	Should		
AC-07.2	US-07	Should		
AC-08.1	US-08	Should		
AC-08.2	US-08	Should		

AC	Story	Priority	Pass / Fail	Notes
AC-09.1	US-09	Must		
AC-09.2	US-09	Must		
AC-09.3	US-09	Must		
AC-10.1	US-10	Must		
AC-10.2	US-10	Must		
AC-10.3	US-10	Must		
AC-11.1	US-11	Should		
AC-11.2	US-11	Should		
AC-11.3	US-11	Should		
AC-11.4	US-11	Should		
AC-12.1	US-12	Could		
AC-12.2	US-12	Could		
AC-13.1	US-13	Should		
AC-13.2	US-13	Should		
AC-13.3	US-13	Should		
AC-14.1	US-14	Should		
AC-14.2	US-14	Should		
AC-14.3	US-14	Should		
AC-14.4	US-14	Should		
AC-15.1	US-15	Could		
AC-15.2	US-15	Could		
AC-15.3	US-15	Could		
AC-15.4	US-15	Could		
AC-16.1	US-16	Could		
AC-16.2	US-16	Could		